Essential Functional Analysis

Prerequisites for the formalisation of Vern Paulsen's lectures on quantum computing

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Abstract

General facts about complex matrices that the formalised lectures (session `Quantum_Computing_by_VP`) build on: sums, adjoints, unitary matrices and Kronecker products, linear-space isomorphisms, and matrices of fixed dimension as a type, with the Hilbert–Schmidt norm (a normed space for the Hahn–Banach theorem of `HOL-Hahn_Banach`) and the structure of a Euclidean space (so that `HOL-Analysis` applies). Nothing here depends on the lectures.

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The gate names of Isabelle_Marries_Dirac would capture the letters X, Y, Z, H, S, T used as variables in the lectures.

hide-const (**open**) *Quantum.X Quantum.Y Quantum.Z Quantum.H Quantum.S Quantum.T*

1 Complex numbers and scalar multiples

lemma *cnj-mult-self-cmod*: *cnj z * z = complex-of-real ((cmod z)²)*
⟨*proof*⟩

lemma *Re-cnj-mult-self*: *Re (cnj z * z) = (cmod z)²*
⟨*proof*⟩

lemma *smult-smult-mat*:
fixes *A* :: *complex Matrix.mat*
shows *a ·m (b ·m A) = (a * b) ·m A*
⟨*proof*⟩

lemma *zero-smult-mat-carrier*:
fixes *A* :: *complex Matrix.mat*
assumes *A ∈ carrier-mat m k*
shows *0 ·m A = 0m m k*
⟨*proof*⟩

2 The adjoint

The adjoint of an $m \times n$ matrix is $n \times m$.

lemma *dagger-carrier*:
assumes *M ∈ carrier-mat m n*
shows *(M†) ∈ carrier-mat n m*
⟨*proof*⟩

lemma *dagger-mult-diag-inner*:
fixes *Y* :: *complex Matrix.mat*
assumes *Yc*: *Y ∈ carrier-mat p q* **and** *j*: *j < q*
shows *((Y†) * Y) $$ (j,j) = ⟨Matrix.col Y j | Matrix.col Y j⟩*
⟨*proof*⟩

3 Sums of vectors

Finite sums of vectors, entrywise.

definition *vec-sum* :: *nat ⇒ 'c set ⇒ ('c ⇒ complex Matrix.vec) ⇒ complex Matrix.vec* **where**
vec-sum n Js w = Matrix.vec n (λi. sum (λJ. w J $ i) Js)

lemma *vec-sum-dim* [*simp*]: *dim-vec (vec-sum n Js w) = n*
⟨*proof*⟩

lemma *vec-sum-index* [*simp*]: *i < n ⟹ vec-sum n Js w $ i = sum (λJ. w J $ i) Js*
⟨*proof*⟩

lemma *vec-sum-cong*:
assumes *⋀J. J ∈ Js ⟹ w J = w' J*
shows *vec-sum n Js w = vec-sum n Js w'*
⟨*proof*⟩

lemma *mult-mat-vec-vec-sum*:
assumes *Ac*: $A \in$ *carrier-mat m n* **and** *fin*: *finite Js*
and *wd*: $\bigwedge J.\ J \in Js \Longrightarrow$ *dim-vec* $(w\ J) = n$
shows $A *_v$ *vec-sum n Js w* $=$ *vec-sum m Js* $(\lambda J.\ A *_v w\ J)$
⟨*proof*⟩

lemma *mult-mat-vec-smult-vec*:
fixes $A ::$ *complex Matrix.mat* **and** $v ::$ *complex Matrix.vec*
assumes *Ac*: $A \in$ *carrier-mat m n* **and** *vd*: *dim-vec* $v = n$
shows $A *_v (c \cdot_v v) = c \cdot_v (A *_v v)$
⟨*proof*⟩

4 Products and sums of matrices

lemma *mat-mult-entry*:
fixes $A\ B ::$ *complex Matrix.mat*
assumes $A \in$ *carrier-mat r s* $B \in$ *carrier-mat s t* $k < r\ j < t$
shows $(A * B)$ \$\$ $(k,j) = (\sum l{<}s.\ A$ \$\$ $(k,l) * B$ \$\$ $(l,j))$
⟨*proof*⟩

The same with the sum over $\{0..<q\}$; square matrices that agree on every vector are equal; a non-zero matrix has a non-zero entry.

lemma *mat-mult-index-sum*:
assumes *A-c*: $A \in$ *carrier-mat p q*
and *B-c*: $B \in$ *carrier-mat q s*
and *r-lt*: $r < p$ **and** *c-lt*: $c < s$
shows $(A * B)$ \$\$ $(r,\ c) = (\sum k = 0..{<}q.\ A$ \$\$ $(r,\ k) * B$ \$\$ $(k,\ c))$
⟨*proof*⟩

lemma *mat-eq-of-mult-vec-eq*:
fixes $M\ N ::$ *complex Matrix.mat*
assumes *M-c*: $M \in$ *carrier-mat k k*
and *N-c*: $N \in$ *carrier-mat k k*
and *eq*: $\bigwedge x.\ x \in$ *carrier-vec* $k \Longrightarrow M *_v x = N *_v x$
shows $M = N$
⟨*proof*⟩

lemma *nonzero-entry*:
fixes $A ::$ *complex Matrix.mat*
assumes *A*: $A \in$ *carrier-mat m k* **and** *nz*: $A \neq 0_m\ m\ k$
obtains $r\ s$ **where** $r < m\ s < k\ A$ \$\$ $(r,s) \neq 0$
⟨*proof*⟩

lemma *foldr-add-mat-carrier*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat n n*
shows *foldr* $(\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ n\ n) \in$ *carrier-mat n n*
⟨*proof*⟩

lemma *foldr-add-mat-index*:
fixes $f :: 'a \Rightarrow$ *complex Matrix.mat*
assumes $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat n n* **and** $i < n$ **and** $j < n$
shows *foldr* $(\lambda x\ acc.\ f\ x + acc)\ xs\ (0_m\ n\ n)$ \$\$ $(i,\ j) =$ *sum-list* $(map\ (\lambda x.\ f\ x$ \$\$ $(i,\ j))\ xs)$
⟨*proof*⟩

lemma *foldr-add-mat-carrier-rect*:
assumes $\forall x \in$ *set xs*. $f\ x \in$ *carrier-mat p q*

shows *foldr* (λx *acc*. *f x* + *acc*) *xs* (0_m *p q*) ∈ *carrier-mat p q*
⟨*proof*⟩

lemma *foldr-add-mat-index-rect*:
assumes ∀ *x* ∈ *set xs*. *f x* ∈ *carrier-mat p q* **and** *i* < *p* **and** *j* < *q*
shows *foldr* (λx *acc*. *f x* + *acc*) *xs* (0_m *p q*) $$ (*i*,*j*) = (∑ *x*←*xs*. *f x* $$ (*i*,*j*))
⟨*proof*⟩

Carrier-preservation in a different dimension pair.

lemma *foldr-add-carrier-dn*:
fixes *f* :: *'a* ⇒ *complex Matrix.mat*
assumes ∀ *x* ∈ *set xs*. *f x* ∈ *carrier-mat d n*
shows *foldr* (λx *acc*. *f x* + *acc*) *xs* (0_m *d n*) ∈ *carrier-mat d n*
⟨*proof*⟩

lemma *foldr-add-mat-cong*:
fixes *f g* :: *'a* ⇒ *complex Matrix.mat*
assumes ⋀*x*. *x* ∈ *set xs* ⟹ *f x* = *g x*
shows *foldr* (λx *acc*. *f x* + *acc*) *xs z* = *foldr* (λx *acc*. *g x* + *acc*) *xs z*
⟨*proof*⟩

Sums over [0.. < *N*], and their entries.

lemma *foldr-upt-add-carrier*:
fixes *f* :: *nat* ⇒ *complex Matrix.mat*
assumes ∀ *k* < *N*. *f k* ∈ *carrier-mat d d*
shows *foldr* (λk *acc*. *f k* + *acc*) [0..<*N*] (0_m *d d*) ∈ *carrier-mat d d*
⟨*proof*⟩

lemma *foldr-upt-add-acc-index*:
fixes *f* :: *nat* ⇒ *complex Matrix.mat*
assumes *carr*: ∀ *k* < *N*. *f k* ∈ *carrier-mat d d*
and *z-c*: *z* ∈ *carrier-mat d d*
and *r-lt*: *r* < *d* **and** *c-lt*: *c* < *d*
shows *foldr* (λk *acc*. *f k* + *acc*) [0..<*N*] *z* $$ (*r*, *c*)
= (∑ *k* = 0..<*N*. *f k* $$ (*r*, *c*)) + *z* $$ (*r*, *c*)
⟨*proof*⟩

lemma *foldr-upt-add-index*:
fixes *f* :: *nat* ⇒ *complex Matrix.mat*
assumes *carr*: ∀ *k* < *N*. *f k* ∈ *carrier-mat d d*
and *r-lt*: *r* < *d* **and** *c-lt*: *c* < *d*
shows *foldr* (λk *acc*. *f k* + *acc*) [0..<*N*] (0_m *d d*) $$ (*r*, *c*)
= (∑ *k* = 0..<*N*. *f k* $$ (*r*, *c*))
⟨*proof*⟩

lemma *foldr-upt-add-acc-index-dn*:
fixes *f* :: *nat* ⇒ *complex Matrix.mat*
assumes *carr*: ∀ *k* < *N*. *f k* ∈ *carrier-mat d n*
and *z-c*: *z* ∈ *carrier-mat d n*
and *r-lt*: *r* < *d* **and** *c-lt*: *c* < *n*
shows *foldr* (λk *acc*. *f k* + *acc*) [0..<*N*] *z* $$ (*r*, *c*)
= (∑ *k* = 0..<*N*. *f k* $$ (*r*, *c*)) + *z* $$ (*r*, *c*)
⟨*proof*⟩

lemma *foldr-smult-nth-index*:
fixes *c* :: *nat* ⇒ *real* **and** *Bs* :: *complex Matrix.mat list*
assumes *Bs-c*: ∀ *B* ∈ *set Bs*. *B* ∈ *carrier-mat n n*
and *r-lt*: *r* < *n* **and** *c-lt*: *c1* < *n*

shows *foldr* (λk *acc. complex-of-real* (*c k*) $\cdot_m$ (*Bs* ! *k*) + *acc*)
[0..<*length Bs*] (0_m *n n*) $$ (*r*, *c1*)
= ($\sum$ *k* = 0..<*length Bs*.
complex-of-real (*c k*) ∗ (*Bs* ! *k*) $$ (*r*, *c1*))
⟨*proof*⟩

lemma *foldr-smult-index*:
assumes *Bs*: ∀ *B* ∈ *set Bs*. *B* ∈ *carrier-mat n m* **and** *r*: *r* < *n* **and** *s*: *s* < *m*
shows *foldr* (λk *acc. c k* $\cdot_m$ (*Bs* ! *k*) + *acc*) [0..<*length Bs*] (0_m *n m*) $$ (*r*,*s*)
= ($\sum$ *k*<*length Bs*. *c k* ∗ *Bs* ! *k* $$ (*r*,*s*))
⟨*proof*⟩

4.1 Algebra of *foldr*-sums of matrices

lemma *foldr-add-mat-mult-left*:
fixes *f* :: *'a* ⇒ *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m n* **and** *fc*: ∀ *x* ∈ *set xs*. *f x* ∈ *carrier-mat n p*
shows *A* ∗ *foldr* (λx *acc. f x* + *acc*) *xs* (0_m *n p*) = *foldr* (λx *acc. A* ∗ *f x* + *acc*) *xs* (0_m *m p*)
⟨*proof*⟩

lemma *foldr-add-mat-mult-right*:
fixes *f* :: *'a* ⇒ *complex Matrix.mat*
assumes *Bc*: *B* ∈ *carrier-mat n p* **and** *fc*: ∀ *x* ∈ *set xs*. *f x* ∈ *carrier-mat m n*
shows *foldr* (λx *acc. f x* + *acc*) *xs* (0_m *m n*) ∗ *B* = *foldr* (λx *acc. f x* ∗ *B* + *acc*) *xs* (0_m *m p*)
⟨*proof*⟩

lemma *foldr-add-mat-dagger*:
fixes *f* :: *'a* ⇒ *complex Matrix.mat*
assumes ∀ *x* ∈ *set xs*. *f x* ∈ *carrier-mat m n*
shows (*foldr* (λx *acc. f x* + *acc*) *xs* (0_m *m n*))† = *foldr* (λx *acc.* (*f x*)† + *acc*) *xs* (0_m *n m*)
⟨*proof*⟩

lemma *foldr-add-mat-zero*:
foldr (λx *acc.* (0_m *m n* :: *complex Matrix.mat*) + *acc*) *xs* (0_m *m n*) = 0_m *m n*
⟨*proof*⟩

lemma *foldr-add-mat-delta*:
fixes *f* :: *nat* ⇒ *complex Matrix.mat*
assumes *fc*: ∀ *x* ∈ *set xs*. *f x* ∈ *carrier-mat m n* **and** *d*: *distinct xs* **and** *i*: *i* ∈ *set xs*
shows *foldr* (λx *acc.* (*if x* = *i then f x else* 0_m *m n*) + *acc*) *xs* (0_m *m n*) = *f i*
⟨*proof*⟩

lemma *foldr-add-mat-smult-pull*:
fixes *f* :: *'a* ⇒ *complex Matrix.mat* **and** *c* :: *'a* ⇒ *complex*
assumes *fc*: ∀ *x* ∈ *set xs*. *f x* = *c x* $\cdot_m$ *M* **and** *Mc*: *M* ∈ *carrier-mat m n*
shows *foldr* (λx *acc. f x* + *acc*) *xs* (0_m *m n*) = *sum-list* (*map c xs*) $\cdot_m$ *M*
⟨*proof*⟩

lemma *foldr-list-as-index*:
fixes *g* :: *complex Matrix.mat* ⇒ *complex Matrix.mat*
shows *foldr* (λA *acc. g A* + *acc*) *As z* = *foldr* (λl *acc. g* (*As* ! *l*) + *acc*) [0..<*length As*] *z*
⟨*proof*⟩

definition *msum* :: *nat* ⇒ *nat* ⇒ (*'a* ⇒ *complex Matrix.mat*) ⇒ *'a list* ⇒ *complex Matrix.mat* **where**
msum r c g xs = *foldr* (λx *acc. g x* + *acc*) *xs* (0_m *r c*)

lemma *msum-Nil* [*simp*]: *msum r c g* [] = 0_m *r c*
⟨*proof*⟩

lemma *msum-Cons* [*simp*]: *msum r c g* (*x* # *xs*) = *g x* + *msum r c g xs*
⟨*proof*⟩

lemma *msum-carrier*: ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c* ⟹ *msum r c g xs* ∈ *carrier-mat r c*
⟨*proof*⟩

lemma *msum-index*:
assumes ∀ *x* ∈ *set xs*. *g x* ∈ *carrier-mat r c* **and** *i* < *r* **and** *j* < *c*
shows *msum r c g xs* $$ (*i*,*j*) = (∑ *x*←*xs*. *g x* $$ (*i*,*j*))
⟨*proof*⟩

5 Finite sums and index arithmetic

lemma *sum-list-product-upt*:
(∑ *x*←*List.product* [*0*..<*a*] [*0*..<*b*]. *h x*) = (∑ *i*<*a*. ∑ *j*<*b*. *h* (*i*,*j*))
⟨*proof*⟩

lemma *sum-delta2*:
fixes *a d b e* :: *nat*
assumes *a*: *a* < *d* **and** *b*: *b* < *e*
shows (∑ *i*<*d*. ∑ *j*<*e*. *if i* = *a* ∧ *j* = *b then v else 0*) = *v*
⟨*proof*⟩

Concatenating rows of equal length, and the reindexing $\kappa = ik + j$.

lemma *concat-uniform-length*:
assumes ∀ *xs* ∈ *set L*. *length xs* = *m*
shows *length* (*concat L*) = *length L* ∗ *m*
⟨*proof*⟩

lemma *concat-uniform-nth*:
assumes *uniform*: ∀ *xs* ∈ *set L*. *length xs* = *m*
and *i-lt*: *i* < *length L*
and *j-lt*: *j* < *m*
shows *concat L* ! (*i* ∗ *m* + *j*) = *L* ! *i* ! *j*
⟨*proof*⟩

lemma *sum-split-prod-atLeastLessThan*:
fixes *f* :: *nat* ⇒ *'a*::*comm-monoid-add*
shows (∑ κ = *0*..<*N* ∗ *k*. *f* κ) = (∑ *i* = *0*..<*N*. ∑ *j* = *0*..<*k*. *f* (*i* ∗ *k* + *j*))
⟨*proof*⟩

lemma *idx8*: *k* < (*8*::*nat*) ⟹ *k* = *0* ∨ *k* = *1* ∨ *k* = *2* ∨ *k* = *3* ∨ *k* = *4* ∨ *k* = *5* ∨ *k* = *6* ∨ *k* = *7*
⟨*proof*⟩

6 Linear maps on a set of matrices

Linearity of a map defined on a set of $n \times n$ matrices (an operator system, from Day 24 on) with values in M_d.

definition *lin-on* :: *nat* ⇒ *nat* ⇒ *complex Matrix.mat set* ⇒ (*complex Matrix.mat* ⇒ *complex Matrix.mat*) ⇒ *bool* **where**
lin-on n d OS Φ ⟷ (∀ *A* ∈ *OS*. Φ *A* ∈ *carrier-mat d d*) ∧ (∀ *A* ∈ *OS*. ∀ *B* ∈ *OS*. Φ (*A* + *B*) = Φ *A* + Φ *B*)
∧ (∀ *A* ∈ *OS*. ∀ *c*. Φ (*c* ·$_m$ *A*) = *c* ·$_m$ Φ *A*)

On a set closed under sums and containing 0, such a map commutes with finite sums.

lemma *msum-closed*:
assumes *S0*: $0_m\ n\ n \in S$ **and** *S*: $\bigwedge A\ B.\ A \in S \Longrightarrow B \in S \Longrightarrow A + B \in S$
and *g*: $\forall x \in set\ xs.\ g\ x \in S$
shows $msum\ n\ n\ g\ xs \in S$
⟨*proof*⟩

lemma *lin-on-msum-closed*:
assumes *L*: *lin-on n d S* Φ **and** *S0*: $0_m\ n\ n \in S$ **and** *S*: $\bigwedge A\ B.\ A \in S \Longrightarrow B \in S \Longrightarrow A + B \in S$
and *g*: $\forall x \in set\ xs.\ g\ x \in S$
shows $\Phi\ (msum\ n\ n\ g\ xs) = msum\ d\ d\ (\lambda x.\ \Phi\ (g\ x))\ xs$
⟨*proof*⟩

7 Linear-space isomorphisms

Matrix spaces are represented by carrier predicates on the single type *complex mat*. The record below packages a carrier with its zero, addition and scalar multiplication, so that statements such as $L(\mathcal{H}_A) \otimes L(\mathcal{H}_B) \cong L(\mathcal{H}_A \otimes \mathcal{H}_B)$ (Day 15) can be made as explicit linear isomorphisms with explicit inverses.

record ('*a*, '*v*) *linear-structure* =
ls-carrier :: '*v set*
ls-zero :: '*v*
ls-add :: '*v* ⇒ '*v* ⇒ '*v*
ls-smul :: '*a* ⇒ '*v* ⇒ '*v*

definition *linear-structure-map* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ ('*v* ⇒ '*w*) ⇒ *bool* **where**
linear-structure-map V W f ≡
f (*ls-zero V*) = *ls-zero W* ∧
(∀ *x* ∈ *ls-carrier V*. *f x* ∈ *ls-carrier W*) ∧
(∀ *x* ∈ *ls-carrier V*. ∀ *y* ∈ *ls-carrier V*.
f (*ls-add V x y*) = *ls-add W* (*f x*) (*f y*)) ∧
(∀ *c x*. *x* ∈ *ls-carrier V* ⟶
f (*ls-smul V c x*) = *ls-smul W c* (*f x*))

definition *is-isomorphism-between* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ ('*v* ⇒ '*w*) ⇒ *bool* **where**
is-isomorphism-between V W f ≡
linear-structure-map V W f ∧
(∃ *g*. *linear-structure-map W V g* ∧
(∀ *x* ∈ *ls-carrier V*. *g* (*f x*) = *x*) ∧
(∀ *y* ∈ *ls-carrier W*. *f* (*g y*) = *y*))

definition *is-isomorphic-to* ::
('*a*, '*v*) *linear-structure* ⇒ ('*a*, '*w*) *linear-structure* ⇒ *bool* **where**
is-isomorphic-to V W ≡ ∃ *f*. *is-isomorphism-between V W f*

lemma *is-isomorphism-betweenD-map*:
assumes *is-isomorphism-between V W f*
shows *linear-structure-map V W f*
⟨*proof*⟩

lemma *is-isomorphism-betweenD-inverse*:
assumes *is-isomorphism-between V W f*
obtains *g* **where** *linear-structure-map W V g*
and $\bigwedge x.\ x \in ls\text{-}carrier\ V \Longrightarrow g\ (f\ x) = x$
and $\bigwedge y.\ y \in ls\text{-}carrier\ W \Longrightarrow f\ (g\ y) = y$

⟨*proof*⟩

The example used throughout: $M_{m,n}$ with entrywise addition and scaling.

definition *mat-linear-structure* :: *nat* ⇒ *nat* ⇒ (*complex*, *complex Matrix.mat*) *linear-structure* **where**
mat-linear-structure m n ≡
(‖ *ls-carrier* = *carrier-mat m n*,
ls-zero = 0_m *m n*,
ls-add = (λ*x y*. *x* + *y*),
ls-smul = (λ*c x*. *c* ·$_m$ *x*) ‖)

hide-const (**open**) *Quantum.X Quantum.Y Quantum.Z Quantum.H Quantum.S Quantum.T*
declare *cauchy-schwarz-eq* [*simp del*] *cauchy-schwarz-col* [*simp del*] *eq-from-inner-prod* [*simp del*]

8 Unitary algebra

Library-level consequences of unitarity ($U^\dagger U = I$ and $UU^\dagger = I$ in the library's definition) used throughout; Paulsen's Day 3 characterisations of unitaries are in theory Day03_Unitary_Matrices.

lemma *unitary-square-dim*:
assumes *uA*: *unitary A*
shows *dim-row A* = *dim-col A*
⟨*proof*⟩

lemma *unitary-inv-l-dimcol*:
assumes *uA*: *unitary A*
shows ($A^\dagger$) ∗ *A* = 1_m (*dim-col A*)
⟨*proof*⟩

lemma *unitary-inv-r-dimrow*:
assumes *uA*: *unitary A*
shows *A* ∗ ($A^\dagger$) = 1_m (*dim-row A*)
⟨*proof*⟩

lemma *unitary-inv-l-dimrow*:
assumes *uA*: *unitary A*
shows ($A^\dagger$) ∗ *A* = 1_m (*dim-row A*)
⟨*proof*⟩

lemma *unitary-inv-r-dimcol*:
assumes *uA*: *unitary A*
shows *A* ∗ ($A^\dagger$) = 1_m (*dim-col A*)
⟨*proof*⟩

lemma *unitary-carrier-square-col*:
assumes *uA*: *unitary A*
shows *A* ∈ *carrier-mat* (*dim-col A*) (*dim-col A*)
⟨*proof*⟩

lemma *unitary-carrier-square-row*:
assumes *uA*: *unitary A*
shows *A* ∈ *carrier-mat* (*dim-row A*) (*dim-row A*)
⟨*proof*⟩

lemma *unitary-dagger*:
assumes *uA*: *unitary A*

shows *unitary* $(A^\dagger)$
⟨*proof*⟩

lemma *unitary-mult*:
assumes *UA*: *unitary A*
and *UB*: *unitary B*
and *dimA*: $A \in$ *carrier-mat n n*
and *dimB*: $B \in$ *carrier-mat n n*
shows *unitary* $(A * B)$
⟨*proof*⟩

lemma *unitary-mult2*:
assumes *UA*: *unitary A*
and *UB*: *unitary B*
and *dims*: *dim-row A* = *dim-row B*
shows *unitary* $(A * B)$
⟨*proof*⟩

9 Kronecker products

Kronecker products of matrices, $A \bigotimes B$: the tensor of identities, the adjoint $(R \otimes T)^* = R^* \otimes T^*$, the mixed-product property $(R_1 \otimes T_1)(R_2 \otimes T_2) = R_1 R_2 \otimes T_1 T_2$ (both from Day 9), and unitarity of tensors of unitaries.

9.1 Tensor of Identities

lemma *tensor-mat-id*:
Tensor.tensor-mat $(1_m\ n)\ (1_m\ m) = 1_m\ (n * m)$
⟨*proof*⟩

9.2 Tensor Adjoint

9.3 Mixed-Product Property

lemma *tensor-mat-mult*:
assumes *dimAC*: *dim-col A* = *dim-row C*
and *dimBD*: *dim-col B* = *dim-row D*
shows *Tensor.tensor-mat A B* ∗ *Tensor.tensor-mat C D* =
Tensor.tensor-mat $(A * C)\ (B * D)$
⟨*proof*⟩

9.4 Corollaries of the Mixed-Product Property

lemma *tensor-adjoint-mult*:
(*Tensor.tensor-mat* $(A^\dagger)\ (B^\dagger)) * ($*Tensor.tensor-mat A B*$) =$
Tensor.tensor-mat $((A^\dagger) * A)\ ((B^\dagger) * B)$
⟨*proof*⟩

lemma *tensor-mult-adjoint*:
(*Tensor.tensor-mat A B*) ∗ (*Tensor.tensor-mat* $(A^\dagger)\ (B^\dagger)) =$
Tensor.tensor-mat $(A * (A^\dagger))\ (B * (B^\dagger))$
⟨*proof*⟩

9.5 Tensor Unitarity

lemma *tensor-mat-unitary-carrier*:
assumes *dimA*: $A \in$ *carrier-mat n n*

and *dimB*: $B \in$ *carrier-mat m m*
and *uA*: *unitary A*
and *uB*: *unitary B*
shows *unitary* (*Tensor.tensor-mat A B*)
⟨*proof*⟩

lemma *tensor-mat-unitary*:
assumes *uA*: *unitary A*
and *uB*: *unitary B*
shows *unitary* (*Tensor.tensor-mat A B*)
⟨*proof*⟩

lemma *tensor-mat-unitary-one-left* [*simp*]:
assumes *uA*: *unitary A*
shows *unitary* (*Tensor.tensor-mat* (1_m *n*) *A*)
⟨*proof*⟩

lemma *tensor-mat-unitary-one-right* [*simp*]:
assumes *uA*: *unitary A*
shows *unitary* (*Tensor.tensor-mat A* (1_m *n*))
⟨*proof*⟩

lemma *one-smult-mat*: (*1*::*complex*) $\cdot_m$ *M* = *M*
⟨*proof*⟩

lemma *mult-index-lt*:
fixes *i j n m* :: *nat*
assumes $i < n$ **and** $j < m$
shows $i * m + j < n * m$
⟨*proof*⟩

lemma *mult-index-div-mod*:
fixes *i l m* :: *nat*
assumes $l < m$
shows (*i* * *m* + *l*) *div m* = *i* (*i* * *m* + *l*) *mod m* = *l*
⟨*proof*⟩

9.6 Tensor products of matrices

lemma *tensor-index-c*:
fixes *A B* :: *complex Matrix.mat*
assumes *A* ∈ *carrier-mat r c B* ∈ *carrier-mat p q i* < *r* * *p j* < *c* * *q* 0 < *c* 0 < *q*
shows *Tensor.tensor-mat A B* $$ (*i*,*j*) = *A* $$ (*i div p*, *j div q*) * *B* $$ (*i mod p*, *j mod q*)
⟨*proof*⟩

lemma *tensor-one-left*:
fixes *A* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m m* **and** *m*: 0 < *m*
shows *Tensor.tensor-mat* (1_m *1*) *A* = *A*
⟨*proof*⟩

lemma *tensor-one-right*:
fixes *A* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat m m* **and** *m*: 0 < *m*
shows *Tensor.tensor-mat A* (1_m *1*) = *A*
⟨*proof*⟩

lemma *tensor-mat-add-right*:
fixes *A B C* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat p q* **and** *Cc*: *C* ∈ *carrier-mat p q* **and** *c*: *0* < *c* **and** *q*: *0* < *q*
shows *Tensor.tensor-mat A* (*B* + *C*) = *Tensor.tensor-mat A B* + *Tensor.tensor-mat A C*
⟨*proof*⟩

lemma *tensor-mat-add-left*:
fixes *A B C* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat r c* **and** *Cc*: *C* ∈ *carrier-mat p q* **and** *c*: *0* < *c* **and** *q*: *0* < *q*
shows *Tensor.tensor-mat* (*A* + *B*) *C* = *Tensor.tensor-mat A C* + *Tensor.tensor-mat B C*
⟨*proof*⟩

lemma *tensor-mat-smult-right*:
fixes *A B* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat p q* **and** *c*: *0* < *c* **and** *q*: *0* < *q*
shows *Tensor.tensor-mat A* (*k* ·$_m$ *B*) = *k* ·$_m$ *Tensor.tensor-mat A B*
⟨*proof*⟩

lemma *tensor-mat-smult-left*:
fixes *A B* :: *complex Matrix.mat*
assumes *Ac*: *A* ∈ *carrier-mat r c* **and** *Bc*: *B* ∈ *carrier-mat p q* **and** *c*: *0* < *c* **and** *q*: *0* < *q*
shows *Tensor.tensor-mat* (*k* ·$_m$ *A*) *B* = *k* ·$_m$ *Tensor.tensor-mat A B*
⟨*proof*⟩

10 Typed-dimension matrix wrapper

The type ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* represents matrices of size *CARD*('*m*) × *CARD*('*n*) with entries in '*a*, by typedef of the JNF matrix type at the appropriate dimension.

typedef (**overloaded**) ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* =
{*M* :: '*a Matrix.mat*. *dim-row M* = *CARD*('*m*) ∧ *dim-col M* = *CARD*('*n*)}
morphisms *unwrap-lm wrap-lm*
⟨*proof*⟩

setup-lifting *type-definition-lmat*

lemma *unwrap-lm-dim-row* [*simp*]:
dim-row (*unwrap-lm* (*M* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)) = *CARD*('*m*)
⟨*proof*⟩

lemma *unwrap-lm-dim-col* [*simp*]:
dim-col (*unwrap-lm* (*M* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)) = *CARD*('*n*)
⟨*proof*⟩

lemma *unwrap-lm-carrier* [*simp*]:
unwrap-lm (*M* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)
∈ *carrier-mat* (*CARD*('*m*)) (*CARD*('*n*))
⟨*proof*⟩

11 Typeclass instances

The unique zero matrix of the right shape.

instantiation *lmat* :: (*zero*, *finite*, *finite*) *zero*
begin
lift-definition *zero-lmat* :: (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
is 0_m (*CARD*(*'m*)) (*CARD*(*'n*)) ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-zero* [*simp*]:
unwrap-lm (*0* :: (*'a* :: *zero*, *'m* :: *finite*, *'n* :: *finite*) *lmat*)
= 0_m (*CARD*(*'m*)) (*CARD*(*'n*))
⟨*proof*⟩

instantiation *lmat* :: (*plus*, *finite*, *finite*) *plus*
begin
lift-definition *plus-lmat* ::
(*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
is (+) ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-plus* [*simp*]:
unwrap-lm ((*A* :: (*'a* :: *plus*, *'m* :: *finite*, *'n* :: *finite*) *lmat*) + *B*)
= *unwrap-lm A* + *unwrap-lm B*
⟨*proof*⟩

instantiation *lmat* :: (*uminus*, *finite*, *finite*) *uminus*
begin
lift-definition *uminus-lmat* ::
(*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
is *uminus* ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-uminus* [*simp*]:
unwrap-lm (− (*A* :: (*'a* :: *uminus*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
= − *unwrap-lm A*
⟨*proof*⟩

instantiation *lmat* :: (*minus*, *finite*, *finite*) *minus*
begin
lift-definition *minus-lmat* ::
(*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ (*'a*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
is (−) ⟨*proof*⟩
instance ⟨*proof*⟩
end

lemma *unwrap-lm-minus* [*simp*]:
unwrap-lm ((*A* :: (*'a* :: *minus*, *'m* :: *finite*, *'n* :: *finite*) *lmat*) − *B*)
= *unwrap-lm A* − *unwrap-lm B*
⟨*proof*⟩

Lift the additive structure laws via ⟦⋀*i j*. ⟦*i* < *dim-row ?B*; *j* < *dim-col ?B*⟧ ⟹ *?A* $$ (*i*, *j*) = *?B* $$ (*i*, *j*); *dim-row ?A* = *dim-row ?B*; *dim-col ?A* = *dim-col ?B*⟧ ⟹ *?A* = *?B* on the underlying JNF matrices.

instance *lmat* :: (*semigroup-add*, *finite*, *finite*) *semigroup-add*
⟨*proof*⟩

instance *lmat* :: (*ab-semigroup-add*, *finite*, *finite*) *ab-semigroup-add*
⟨*proof*⟩

instance *lmat* :: (*monoid-add*, *finite*, *finite*) *monoid-add*
⟨*proof*⟩

instance *lmat* :: (*comm-monoid-add*, *finite*, *finite*) *comm-monoid-add*
⟨*proof*⟩

instance *lmat* :: (*group-add*, *finite*, *finite*) *group-add*
⟨*proof*⟩

instance *lmat* :: (*ab-group-add*, *finite*, *finite*) *ab-group-add*
⟨*proof*⟩

12 Real scalar multiplication and Hahn-Banach overloading

HOL's $(\cdot)$ is the abstract real-scalar multiplication used by the Hahn-Banach locales. We overload it on the type (*complex*, *'m*, *'n*) *lmat* as multiplication by the embedded real, lifted from JNF's $(\cdot_m)$.

overloading
prod-lmat ≡
Vector-Space.prod ::
real ⇒ (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
⇒ (*complex*, *'m*, *'n*) *lmat*
begin
definition *prod-lmat* ::
real ⇒ (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*
⇒ (*complex*, *'m*, *'n*) *lmat*
where
prod-lmat r M = *wrap-lm* (*complex-of-real r* $\cdot_m$ *unwrap-lm M*)
end

lemma *unwrap-lm-prod* [*simp*]:
unwrap-lm (*Vector-Space.prod r* (*M* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat*))
= *complex-of-real r* $\cdot_m$ *unwrap-lm M*
⟨*proof*⟩

13 Real-vectorspace locale on the full type

Every (*complex*, *'m*, *'n*) *lmat* with the universe set is a real vector space in HOL/Hahn_Banach's sense.

lemma *vectorspace-UNIV-lmat*:
Vector-Space.vectorspace (*UNIV* :: (*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat set*)
⟨*proof*⟩

14 The Hilbert-Schmidt norm on (*complex*, *'m*, *'n*) *lmat*

The Hilbert–Schmidt (Frobenius) norm $\|M\| = (\sum_{i,j} |M_{ij}|^2)^{1/2}$, the Euclidean norm of the entries.

definition *hs-norm-lmat* ::
(*complex*, *'m* :: *finite*, *'n* :: *finite*) *lmat* ⇒ *real*

where
hs-norm-lmat M =
L2-set (λ(*i*, *j*). *cmod* (*unwrap-lm M* $$ (*i*, *j*))) ({*0*..<*CARD*('*m*)} × {*0*..<*CARD*('*n*)})

lemma *hs-norm-lmat-nonneg*: *0* ≤ *hs-norm-lmat M*
⟨*proof*⟩

Scaling every entry in modulus by $r \geq 0$ scales the norm by r.

lemma *hs-norm-lmat-scale*:
fixes *M N* :: (*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes ⋀*i j*. *i* < *CARD*('*m*) ⟹ *j* < *CARD*('*n*) ⟹
cmod (*unwrap-lm N* $$ (*i*, *j*)) = *r* * *cmod* (*unwrap-lm M* $$ (*i*, *j*))
and *0* ≤ *r*
shows *hs-norm-lmat N* = *r* * *hs-norm-lmat M*
⟨*proof*⟩

lemma *hs-norm-lmat-smult*:
hs-norm-lmat (*Vector-Space.prod a M*) = |*a*| * *hs-norm-lmat M*
⟨*proof*⟩

lemma *hs-norm-lmat-triangle*:
fixes *M N* :: (*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
shows *hs-norm-lmat* (*M* + *N*) ≤ *hs-norm-lmat M* + *hs-norm-lmat N*
⟨*proof*⟩

15 Subspace, seminorm, normed-vectorspace instances

HS norm is a seminorm on the whole real vector space of (*complex*, '*m*, '*n*) *lmat*.

lemma *seminorm-UNIV-hs-norm-lmat*:
seminorm (*UNIV* :: (*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat set*) *hs-norm-lmat*
⟨*proof*⟩

HS norm is a (real) norm: it vanishes only on the zero matrix.

lemma *hs-norm-lmat-zero-iff*:
fixes *M* :: (*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
shows (*hs-norm-lmat M* = *0*) ⟷ (*M* = *0*)
⟨*proof*⟩

lemma *norm-UNIV-hs-norm-lmat*:
Normed-Space.norm
(*UNIV* :: (*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat set*) *hs-norm-lmat*
⟨*proof*⟩

lemma *normed-vectorspace-UNIV-lmat*:
normed-vectorspace
(*UNIV* :: (*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat set*) *hs-norm-lmat*
⟨*proof*⟩

A complex-linear subspace V of (*complex*, '*m*, '*n*) *lmat* (i.e. nonempty, closed under addition and complex-scalar multiplication) is in particular a real-linear subspace, hence a *subspace* of *UNIV* in HOL/Hahn_Banach's sense.

definition *cmat-subspace-lmat* ::
(*complex*, '*m* :: *finite*, '*n* :: *finite*) *lmat set* ⇒ *bool*
where
cmat-subspace-lmat V ≡
V ≠ {} ∧

$(\forall x \in V.\ \forall y \in V.\ x + y \in V) \wedge$
$(\forall x \in V.\ \forall c :: complex.$
$wrap\text{-}lm\ (c \cdot_m unwrap\text{-}lm\ x) \in V)$

lemma *subspace-of-cmat-subspace-lmat*:
fixes $V :: (complex, 'm :: finite, 'n :: finite)\ lmat\ set$
assumes *cmat-subspace-lmat V*
shows *Subspace.subspace V* $(UNIV :: (complex, 'm, 'n)\ lmat\ set)$
⟨*proof*⟩

16 Complex scalar multiplication on lmat

Complex scalar multiplication on $(complex, 'm, 'n)\ lmat$, extending the *Vector-Space.prod* overload (which is only by reals).

definition *cmult-lmat* ::
$complex \Rightarrow (complex, 'm :: finite, 'n :: finite)\ lmat$
$\Rightarrow (complex, 'm, 'n)\ lmat$
where
$cmult\text{-}lmat\ c\ M = wrap\text{-}lm\ (c \cdot_m unwrap\text{-}lm\ M)$

lemma *unwrap-lm-cmult* [*simp*]:
unwrap-lm $(cmult\text{-}lmat\ c\ (M :: (complex, 'm :: finite, 'n :: finite)\ lmat))$
$= c \cdot_m unwrap\text{-}lm\ M$
⟨*proof*⟩

lemma *cmult-lmat-real*:
$cmult\text{-}lmat\ (complex\text{-}of\text{-}real\ a)\ M = Vector\text{-}Space.prod\ a\ M$
⟨*proof*⟩

lemma *cmult-lmat-add*:
fixes $x\ y :: (complex, 'm :: finite, 'n :: finite)\ lmat$
shows $cmult\text{-}lmat\ c\ (x + y) = cmult\text{-}lmat\ c\ x + cmult\text{-}lmat\ c\ y$
⟨*proof*⟩

lemma *cmult-lmat-scalar-add*:
fixes $x :: (complex, 'm :: finite, 'n :: finite)\ lmat$
shows $cmult\text{-}lmat\ (c + d)\ x = cmult\text{-}lmat\ c\ x + cmult\text{-}lmat\ d\ x$
⟨*proof*⟩

lemma *cmult-lmat-scalar-mult*:
fixes $x :: (complex, 'm :: finite, 'n :: finite)\ lmat$
shows $cmult\text{-}lmat\ (c * d)\ x = cmult\text{-}lmat\ c\ (cmult\text{-}lmat\ d\ x)$
⟨*proof*⟩

lemma *cmult-lmat-one*:
fixes $x :: (complex, 'm :: finite, 'n :: finite)\ lmat$
shows $cmult\text{-}lmat\ 1\ x = x$
⟨*proof*⟩

lemma *cmult-lmat-uminus-one*:
fixes $x :: (complex, 'm :: finite, 'n :: finite)\ lmat$
shows $cmult\text{-}lmat\ (-\ 1)\ x = -\ x$
⟨*proof*⟩

lemma *cmult-lmat-i-sq*:
fixes $x :: (complex, 'm :: finite, 'n :: finite)\ lmat$
shows $cmult\text{-}lmat\ \mathrm{i}\ (cmult\text{-}lmat\ \mathrm{i}\ x) = -\ x$

⟨*proof*⟩

lemma *hs-norm-lmat-cmult*:
fixes M :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *hs-norm-lmat* (*cmult-lmat c M*) = *cmod c* ∗ *hs-norm-lmat M*
⟨*proof*⟩

lemma *hs-norm-lmat-cmult-i* [*simp*]:
fixes M :: (*complex*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
shows *hs-norm-lmat* (*cmult-lmat* i *M*) = *hs-norm-lmat M*
⟨*proof*⟩

The HOL/Hahn_Banach development imports locales also called *norm*, *seminorm* and *continuous*. Hide their short names so that *norm* is the class method from *HOL.Real-Vector-Spaces* and *continuous* the topological predicate.

hide-const (**open**) *Normed-Space.norm*
hide-const (**open**) *Normed-Space.seminorm*
hide-const (**open**) *Function-Norm.continuous*

17 Overview

Nothing in this theory is stated in the notes; it is the bridge that lets the HOL-Analysis theorems on convex sets (separating hyperplanes, compactness) be applied to matrices. ($'a$, $'m$, $'n$) *lmat* is given the typeclass tower *real-vector*, *metric-space*, *real-normed-vector*, *real-inner*, *euclidean-space*, *heine-borel*; for $'a$ = *complex* the norm is the Hilbert–Schmidt norm, the inner product its real part, and the dimension is $2\,\mathrm{CARD}('m)\,\mathrm{CARD}('n)$.

18 Real scalar multiplication and *real-vector* instance

For $'a$ :: *real-vector*, lift element-wise scalar multiplication via *map-mat* (*scaleR a*).

instantiation *lmat* :: (*real-vector*, *finite*, *finite*) *scaleR*
begin

lift-definition *scaleR-lmat* ::
real ⇒ ($'a$, $'m$ :: *finite*, $'n$ :: *finite*) *lmat* ⇒ ($'a$, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
is λ*r M*. *map-mat* (λ*x*. *r* ∗$_R$ *x*) *M* ⟨*proof*⟩

instance ⟨*proof*⟩

end

lemma *unwrap-lm-scaleR* [*simp*]:
unwrap-lm ((*r* :: *real*) ∗$_R$ (*M* :: ($'a$ :: *real-vector*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*))
= *map-mat* (λ*x*. *r* ∗$_R$ *x*) (*unwrap-lm M*)
⟨*proof*⟩

lemma *unwrap-lm-scaleR-index* [*simp*]:
fixes M :: ($'a$ :: *real-vector*, $'m$ :: *finite*, $'n$ :: *finite*) *lmat*
assumes $i < CARD('m)$ $j < CARD('n)$
shows *unwrap-lm* (*r* ∗$_R$ *M*) $$ (*i*, *j*) = *r* ∗$_R$ (*unwrap-lm M* $$ (*i*, *j*))
⟨*proof*⟩

instance *lmat* :: (*real-vector*, *finite*, *finite*) *real-vector*
⟨*proof*⟩

19 Entries of sums and differences

lemma *unwrap-lm-plus-index* [*simp*]:
fixes $M\ N :: ('a :: plus, 'm :: finite, 'n :: finite)\ lmat$
assumes $i < CARD('m)\ j < CARD('n)$
shows $unwrap\text{-}lm\ (M + N)\ \$\$\ (i, j) = unwrap\text{-}lm\ M\ \$\$\ (i, j) + unwrap\text{-}lm\ N\ \$\$\ (i, j)$
⟨*proof*⟩

lemma *unwrap-lm-minus-index* [*simp*]:
fixes $M\ N :: ('a :: ab\text{-}group\text{-}add, 'm :: finite, 'n :: finite)\ lmat$
assumes $i < CARD('m)\ j < CARD('n)$
shows $unwrap\text{-}lm\ (M - N)\ \$\$\ (i, j) = unwrap\text{-}lm\ M\ \$\$\ (i, j) - unwrap\text{-}lm\ N\ \$\$\ (i, j)$
⟨*proof*⟩

lemma *unwrap-lm-zero-index* [*simp*]:
fixes $i\ j :: nat$
assumes $i < CARD('m :: finite)\ j < CARD('n :: finite)$
shows $unwrap\text{-}lm\ (0 :: ('a :: zero, 'm, 'n)\ lmat)\ \$\$\ (i, j) = 0$
⟨*proof*⟩

lemma *lmat-eq-iff-entries*:
fixes $M\ N :: ('a, 'm :: finite, 'n :: finite)\ lmat$
shows $M = N \longleftrightarrow$
$(\forall\, i\ j.\ i < CARD('m) \longrightarrow j < CARD('n)$
$\longrightarrow unwrap\text{-}lm\ M\ \$\$\ (i, j) = unwrap\text{-}lm\ N\ \$\$\ (i, j))$
⟨*proof*⟩

20 Distance, uniformity, open: *metric-space* instance

Distance is the L2 sum of entry-wise distances, indexed by $\{0..<CARD('m)\} \times \{0..<CARD('n)\}$. Uniformity and *open* are defined via the standard metric-induced filter so that the *open-uniformity* axiom is discharged by reflexivity.

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *dist*
begin

definition *dist-lmat* :: $('a, 'm :: finite, 'n :: finite)\ lmat$
$\Rightarrow ('a, 'm :: finite, 'n :: finite)\ lmat \Rightarrow real$ **where**
$dist\text{-}lmat\ M\ N = L2\text{-}set\ (\lambda(i, j).\ dist\ (unwrap\text{-}lm\ M\ \$\$\ (i, j))\ (unwrap\text{-}lm\ N\ \$\$\ (i, j)))$
$(\{0..<CARD('m)\} \times \{0..<CARD('n)\})$

instance ⟨*proof*⟩

end

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *uniformity-dist*
begin

definition *uniformity-lmat* ::
$(('a, 'm :: finite, 'n :: finite)\ lmat \times ('a, 'm, 'n)\ lmat)\ filter$ **where**
$(uniformity\text{-}lmat :: (('a, 'm, 'n)\ lmat \times ('a, 'm, 'n)\ lmat)\ filter)$
$= (INF\ e \in \{0<..\}.\ principal\ \{(x, y).\ dist\ x\ y < e\})$

instance ⟨*proof*⟩

end

instantiation *lmat* :: (*metric-space*, *finite*, *finite*) *metric-space*

begin

definition *open-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat set* ⇒ *bool* **where**
open-lmat U = (∀ *x* ∈ *U*. ∀ *F* (*x'*, *y*) *in uniformity*. *x'* = *x* ⟶ *y* ∈ *U*)

instance ⟨*proof*⟩

end

21 Norm and *real-normed-vector* instance

instantiation *lmat* :: (*real-normed-vector*, *finite*, *finite*) *real-normed-vector*
begin

definition *norm-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ *real* **where**
norm-lmat M = *L2-set* (λ(*i*, *j*). *norm* (*unwrap-lm M* $$ (*i*, *j*)))
({*0*..<*CARD*('*m*)} × {*0*..<*CARD*('*n*)})

definition *sgn-lmat* ::
('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* ⇒ ('*a*, '*m*, '*n*) *lmat* **where**
sgn-lmat (*M* :: ('*a*, '*m*, '*n*) *lmat*) = *scaleR* (*inverse* (*norm M*)) *M*

instance ⟨*proof*⟩

end

22 Real inner product

For '*a* :: *real-inner*, the lmat inner product is the entry-wise sum of inner products. When '*a* = *complex* with the standard real-inner instance *inner z w* = *Re* (*cnj z* ∗ *w*), this is the real part of the Hilbert-Schmidt inner product.

instantiation *lmat* :: (*real-inner*, *finite*, *finite*) *real-inner*
begin

definition *inner-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
⇒ ('*a*, '*m*, '*n*) *lmat* ⇒ *real* **where**
inner-lmat M N = (∑ (*i*, *j*) ∈ {*0*..<*CARD*('*m*)} × {*0*..<*CARD*('*n*)}.
inner (*unwrap-lm M* $$ (*i*, *j*)) (*unwrap-lm N* $$ (*i*, *j*)))

instance ⟨*proof*⟩

end

23 Basis: matrix units carrying a basis vector

definition *lmat-axis* :: *nat* ⇒ *nat* ⇒ '*a* :: *zero*
⇒ ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat* **where**
lmat-axis i j u = *wrap-lm* (*Matrix.mat* (*CARD*('*m*)) (*CARD*('*n*))
(λ(*r*, *s*). *if r* = *i* ∧ *s* = *j then u else 0*))

lemma *unwrap-lmat-axis* [*simp*]:
unwrap-lm (*lmat-axis i j u* :: ('*a* :: *zero*, '*m* :: *finite*, '*n* :: *finite*) *lmat*)
= *Matrix.mat* (*CARD*('*m*)) (*CARD*('*n*))
(λ(*r*, *s*). *if r* = *i* ∧ *s* = *j then u else 0*)
⟨*proof*⟩

lemma *lmat-axis-index* [*simp*]:
fixes $u :: 'a :: zero$
assumes $r < CARD('m :: finite)$ $s < CARD('n :: finite)$
shows *unwrap-lm* (*lmat-axis i j u* :: $('a, 'm, 'n)$ *lmat*) $$ (r, s)
$= (if\ r = i \wedge s = j\ then\ u\ else\ 0)$
⟨*proof*⟩

Inner product against a single matrix-unit basis vector picks out one matrix entry.

lemma *inner-lmat-axis*:
fixes $u :: 'a :: real\text{-}inner$
and $M :: ('a, 'm :: finite, 'n :: finite)$ *lmat*
assumes *ij*: $i < CARD('m)$ $j < CARD('n)$
shows *inner M* (*lmat-axis i j u* :: $('a, 'm, 'n)$ *lmat*)
$=$ *inner* (*unwrap-lm M* $$ (i, j)) u
⟨*proof*⟩

Two matrix-unit basis vectors are equal iff their entries and positions match (or both vectors are zero).

lemma *lmat-axis-eq-iff*:
fixes $u\ v :: 'a :: zero$
assumes $i < CARD('m :: finite)$ $j < CARD('n :: finite)$
and $i' < CARD('m)$ $j' < CARD('n)$
shows (*lmat-axis i j u* :: $('a, 'm, 'n)$ *lmat*) = *lmat-axis* $i'\ j'\ v$
$\longleftrightarrow (u = v \wedge i = i' \wedge j = j') \vee (u = 0 \wedge v = 0)$
⟨*proof*⟩

Inner product between two matrix-unit basis vectors: 1 if same position and basis component, 0 otherwise.

lemma *inner-lmat-axis-axis*:
fixes $u\ v :: 'a :: real\text{-}inner$
assumes *ij*: $i < CARD('m :: finite)$ $j < CARD('n :: finite)$
and *ij'*: $i' < CARD('m)$ $j' < CARD('n)$
shows *inner* (*lmat-axis i j u* :: $('a, 'm, 'n)$ *lmat*) (*lmat-axis* $i'\ j'\ v$)
$= (if\ i = i' \wedge j = j'\ then\ inner\ u\ v\ else\ 0)$
⟨*proof*⟩

24 Basis set

definition *Basis-lmat-set* :: $('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)$ *lmat set*
where *Basis-lmat-set* $= (\bigcup i \in \{0..<CARD('m)\}.$
$\bigcup j \in \{0..<CARD('n)\}.$
$\bigcup u \in Basis.\ \{lmat\text{-}axis\ i\ j\ u :: ('a, 'm, 'n)\ lmat\})$

lemma *Basis-lmat-set-finite*:
finite (*Basis-lmat-set* :: $('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)$ *lmat set*)
⟨*proof*⟩

lemma *Basis-lmat-set-nonempty*:
Basis-lmat-set $\neq (\{\} :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat\ set)$
⟨*proof*⟩

lemma *Basis-lmat-orthonormal*:
fixes $u\ v :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)$ *lmat*
assumes *uB*: $u \in$ *Basis-lmat-set* **and** *vB*: $v \in$ *Basis-lmat-set*
shows *inner u v* $= (if\ u = v\ then\ 1\ else\ 0)$
⟨*proof*⟩

lemma *axis-mem-Basis-lmat-set*:
fixes $u :: 'a :: euclidean\text{-}space$
assumes *i-lt*: $i < CARD('m :: finite)$
and *j-lt*: $j < CARD('n :: finite)$
and *uB*: $u \in Basis$
shows $(lmat\text{-}axis\ i\ j\ u :: ('a, 'm, 'n)\ lmat) \in Basis\text{-}lmat\text{-}set$
⟨*proof*⟩

lemma *Basis-lmat-perp-imp-zero*:
fixes $x :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
assumes $*$: $\forall u \in Basis\text{-}lmat\text{-}set.\ inner\ x\ u = 0$
shows $x = 0$
⟨*proof*⟩

lemma *zero-inner-Basis-lmat*:
fixes $u :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
shows $inner\ (0 :: ('a, 'm, 'n)\ lmat)\ u = 0$
⟨*proof*⟩

lemma *Basis-lmat-perp-zero*:
fixes $x :: ('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat$
shows $(\forall u \in Basis\text{-}lmat\text{-}set.\ inner\ x\ u = 0) \longleftrightarrow x = 0$
⟨*proof*⟩

25 Euclidean space instance

instantiation *lmat* :: (*euclidean-space*, *finite*, *finite*) *euclidean-space*
begin

definition *Basis-lmat* :: ('*a*, '*m* :: *finite*, '*n* :: *finite*) *lmat set* **where**
Basis-lmat = *Basis-lmat-set*

instance ⟨*proof*⟩

end

26 Heine-Borel

instance *lmat* :: (*euclidean-space*, *finite*, *finite*) *heine-borel* ⟨*proof*⟩

27 Dimension

Two distinct (i, j, u) with $u \in Basis\text{-}'a$ give distinct matrix-axis vectors.

lemma *lmat-axis-inj-on-Basis-indices*:
fixes $u\ v :: 'a :: euclidean\text{-}space$
assumes *ij*: $i < CARD('m :: finite)\ j < CARD('n :: finite)$
and *ij'*: $i' < CARD('m)\ j' < CARD('n)$
and *uB*: $u \in Basis$ **and** *vB*: $v \in Basis$
shows $(lmat\text{-}axis\ i\ j\ u :: ('a, 'm, 'n)\ lmat) = lmat\text{-}axis\ i'\ j'\ v$
$\longleftrightarrow i = i' \wedge j = j' \wedge u = v$
⟨*proof*⟩

lemma *DIM-lmat*:
$DIM(('a :: euclidean\text{-}space, 'm :: finite, 'n :: finite)\ lmat)$
$= CARD('m) * CARD('n) * DIM('a)$

⟨*proof*⟩

28 Component-wise bounds via the L2 norm

lemma *norm-lmat-index-le*:
fixes M :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes $i < CARD('m)$ $j < CARD('n)$
shows *norm* (*unwrap-lm* M $$ $(i, j)) \leq$ *norm* M
⟨*proof*⟩

lemma *dist-lmat-index-le*:
fixes M N :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes $i < CARD('m)$ $j < CARD('n)$
shows *dist* (*unwrap-lm* M $$ $(i, j))$ (*unwrap-lm* N $$ $(i, j)) \leq$ *dist* M N
⟨*proof*⟩

29 Convergence: lmat sequence convergence implies entry convergence

lemma *tendsto-unwrap-lm-index*:
fixes f :: '*b* ⇒ ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
and L :: ('*a*, '*m*, '*n*) *lmat*
assumes *lim*: $(f \longrightarrow L)$ F
and *ij*: $i < CARD('m)$ $j < CARD('n)$
shows $((\lambda x.$ *unwrap-lm* $(f\ x)$ $$ $(i, j)) \longrightarrow$ *unwrap-lm* L $$ $(i, j))$ F
⟨*proof*⟩

lemma *LIMSEQ-unwrap-lm-index*:
fixes f :: *nat* ⇒ ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
and L :: ('*a*, '*m*, '*n*) *lmat*
assumes $f \longrightarrow L$ $i < CARD('m)$ $j < CARD('n)$
shows $(\lambda n.$ *unwrap-lm* $(f\ n)$ $$ $(i, j)) \longrightarrow$ *unwrap-lm* L $$ (i, j)
⟨*proof*⟩

30 Entries as continuous functions; wrapping

The entry-projection $M \mapsto$ *unwrap-lm* M $$ (i, j) is continuous on lmat for valid index (i, j).

lemma *isCont-unwrap-lm-index*:
fixes x :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat*
assumes *ij*: $i < CARD('m)$ $j < CARD('n)$
shows *isCont* $(\lambda M.$ *unwrap-lm* M $$ $(i, j))$ x
⟨*proof*⟩

lemma *continuous-on-unwrap-lm-index*:
fixes S :: ('*a* :: *real-normed-vector*, '*m* :: *finite*, '*n* :: *finite*) *lmat set*
assumes *ij*: $i < CARD('m)$ $j < CARD('n)$
shows *continuous-on* S $(\lambda M.$ *unwrap-lm* M $$ $(i, j))$
⟨*proof*⟩

lemma *unwrap-wrap-lm*:
fixes M :: *complex Matrix.mat*
assumes $M \in$ *carrier-mat* $CARD('k::finite)$ $CARD('k)$
shows *unwrap-lm* (*wrap-lm* M :: (*complex*, '*k*, '*k*) *lmat*) $= M$
⟨*proof*⟩

lemma *unwrap-lm-scaleR-mat*:
fixes R :: (*complex*, $'k$ :: *finite*, $'k$) *lmat*
shows *unwrap-lm* (u $*_R$ R) = *complex-of-real* u $\cdot_m$ *unwrap-lm* R
⟨*proof*⟩